Reflection questions

1)Why is it useful to specify action preconditions and effects before asking an LLM to write the planner?

ANS- It is useful because then the LLM will have more context to work on and generate and curate the code to a certain specificity.

2)Give an example of an error that could occur if the planner failed to check an action's preconditions.

ANS-If the planner does not check preconditions, it might allow the robot to pick up the package even when the robot and package are at different locations. For example, the robot could be at B while the package is at A, but the planner might still execute pickup{Package,A}. This would result in an invalid state.

3)Why is a plan that "looks reasonable" not necessarily a valid plan?

ANS- It could be because plan may look reasonable but violate an action's preconditions or produce an incorrect state. For example, the robot might try to move through a wall or drop the package without holding it. A valid plan must satisfy all action preconditions and reach the goal through correctly applied effects.

4). What did the LLM contribute to the implementation?

ANS- The LLM contributed in generating the code from the planner I gave it and adding a few more details to curate the prompt for the generation.It also defined the action class.

5)What did you have to verify independently?

ANS-I had to verify that the warehouse grid was represented correctly, movement actions did not cross walls, preconditions and effects were implemented correctly, and the resulting plan reached the goal. I also needed to check that BFS detected when no valid plan existed.

6)In this laboratory, where is logical reasoning being used?

ANS-Logical Reasoning is being used to check whether the condition and preconditions have been satisifed before executing it and updates the state using its effects

7)How is planning related to the search algorithms studied in the previous module?

ANS-Planning can be formulated as a search problem in which each state is a node and each applicable action is an edge connecting states. BFS explores these states systematically until it finds one satisfying the goal.